



Empirical Methods for Finance

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Graded Group Assignment: Attention-induced Trading among Robinhood Investors

GROUP A31:

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Sample selection and summary statistics. Get to know the data!

1. Once the observations with negative/missing prices and those corresponding to the Robinhood outages on March 2, March 3, and June 18, 2020, are dropped, the sample includes 3,817 distinct stocks and 544 distinct days.
2.
 - a. The table below reports the yearly summary statistics for the rh variable:

year	N	Mean	SD	p50
2018	470477	1758.96	9945.933	205
2019	787498	2452.887	13544.21	285
2020	492310	5663.198	31400.03	545
Total	1750285	3169.337	19722.25	307

Table 1

- b. By looking at the summary statistics, it can be concluded that statements iii. and iv. are correct. In 2020, the median number of investors holding a specific stock is 545, as highlighted by the green circle in the table above, and the dataset is indeed structured at the stock-day level, meaning that each observation corresponds to a listed stock on a given day.
3. The following table summarizes a distribution of four key statistics: the average value is 510.87; the median is 545; the minimum observed value is 2; the maximum observed value is 545:

Avg	Med	Min	Max
510.8692	545	2	545

Table 2: avg, median, min, max of observations

Not all tickers are observed for the same number of days because some stocks are not traded throughout the entire sample period. This can happen due to IPOs, delistings, mergers, or acquisitions. In the case of Sprint

Corporation (ticker: S), we only observe 455 days since the company merged with T-Mobile and was subsequently delisted, meaning it was no longer traded as a separate stock.

Top mover list.

4.
 - a. Daily market capitalization is calculated as the stock price multiplied by shares outstanding (already in thousands), divided by 1,000 to express the result in millions of dollars. We expressed the formula using the considered variable in the code.
 - b. See code.
 - c. See code.
 - d. See code.
5.
 - a. See code.
 - b. See code.
 - c. The table indicates that 67.38% of the observations classified as top movers correspond to top gainers (i.e., cases where `pos_ret` equals 1), while only 32.62% are top losers:

posret	Freq.	Percent	Cum.
0	3,549	32.62	32.62
1	7,331	67.38	100.00
Total	10,880	100.00	

Table 3: ratio of gaining and losing top movers

- d. The table provides a statistical summary of various financial metrics, categorized by the `topmover` dummy variable (0 for non-top movers and 1 for top movers). For each category, it displays the number of observations (`n`), mean, standard deviation (`sd`), and median for variables including `mktpcap` (market capitalization), `ret` (return), `abs_ret` (absolute return), `userchg` (user change), and `userra_o` (user ratio):

topmover	mktcapt	ret	abs_ret	userchg	userratio
0	1735588 8665.328 42955.42 851.4995	1735588 .0000436 .0384479 0	1735588 .0240791 .0299739 .0145345	1735588 13.59478 317.2785 0	1730107 1.006484 .3467726 1
1	10880 845.0439 6035.425 86.28014	10880 .2332419 3.637141 .1823401	10880 .3888794 3.623804 .2321107	10880 268.9728 1642.033 9	10778 1.219268 2.005789 1.015814
Total	1746468 8616.61 42828.48 840.1722	1746468 .0014964 .2901892 0	1746468 .0263517 .2889941 .0146721	1746468 15.18571 342.4 0	1740885 1.007801 .3803828 1

Table 4: mean, standard deviation, median of top movers mktcap, ret, abs_ret, userchg, userratio

e. See code.

f. After running the regression of *userchg* on *lag_topmover* we obtain the following estimated model:

$$\widehat{userchg} = 12.3105 + 420.3837lag_topmover$$

. reg userchg lag_topmover

Source	SS	df	MS	Number of obs	=	1,742,651
Model	1.9039e+09	1	1.9039e+09	F(1, 1742649)	=	17826.45
Residual	1.8612e+11	1,742,649	106803.633	Prob > F	=	0.0000
				R-squared	=	0.0101
				Adj R-squared	=	0.0101
Total	1.8803e+11	1,742,650	107896.12	Root MSE	=	326.81

userchg	Coefficient	Std. err.	t	P> t	[95% conf. interval]
lag_topmover	420.3837	3.148571	133.52	0.000	414.2126 426.5547
_cons	12.31046	.2483378	49.57	0.000	11.82373 12.7972

Table 5: regression userchg on lag_topmover

i. $\widehat{\beta}_0 = 12.3105$

The intercept β_0 represents the expected daily change in the number of Robinhood investors when a stock was not featured as a top mover on the previous trading day. On average, such ticker experiences an inflow of approximately 12 new users per day.

ii. $\widehat{\beta}_1 = 420.3837$

The slope coefficient captures the incremental effect of having been a top mover in the prior day. Its

positive magnitude suggests that, on average, tickers appearing on the top movers list attract an additional 420 Robinhood users on the following day compared to non-top movers. This indicates that ticker identified as top mover attract substantially more attention from investors, leading to a significant increase in user engagement the following day.

iii.

A. ***P – value* = 0.000**

This value is far below the conventional significance level of 0.01, 0.05 and 0.10. This mean that there is a statistically significant relationship between a *topmover* occurring the previous day and the *userchg* variable.

B. ***T – statistic* = 133.5200**

This value is extremely high. A very large t-statistic like this indicates that the coefficient of 420.38 is many standard errors away from zero, providing strong evidence that the true coefficient is not zero. This confirms the statistical significance of the result.

C. ***Confidence Interval* = [414.2126, 426.5547]**

Since this entire interval is positive and does not contain zero, we are 95% confident that the true effect of a *topmover* on *userchg* is a positive increase, somewhere between 414.21 and 426.55.

This also confirms the statistical significance of the coefficient.

- g. This model examines the effect of a *topmover* event on the user ratio. This ratio is likely the daily user count divided by the previous day's user count, representing the growth factor. A value of 1.0 means that there's no change, while having a value greater than 1.0 indicates the presence of growth. After running the regression of *userratio* on *lag_topmover* we obtain the following estimated model:

$$\widehat{userratio} = 1.0052 + 0.2245lag_topmover$$

. reg userratio lag_topmover

Source	SS	df	MS	Number of obs	=	1,737,406
Model	538.560925	1	538.560925	F(1, 1737404)	=	8359.98
Residual	111925.812	1,737,404	.064421293	Prob > F	=	0.0000
				R-squared	=	0.0048
				Adj R-squared	=	0.0048
Total	112464.373	1,737,405	.064731236	Root MSE	=	.25381

userratio	Coefficient	Std. err.	t	P> t	[95% conf. interval]
lag_topmover	.2245127	.0024555	91.43	0.000	.2197 .2293253
_cons	1.005283	.0001932	5204.47	0.000	1.004904 1.005661

Table 6: regression *userratio* on *lag_topmover*

$$\widehat{\beta}_0 = 1.0052$$

This is the average userratio on days that do not follow a topmover. A value slightly above 1.0 indicates that, on a typical day, the user count increases by a small amount (about 0.53%).

$$\widehat{\beta}_1 = 0.2245$$

This indicates that, on average, a day following a topmover event has a user ratio that is 0.22 units higher than a day without a topmover. This effect is highly significant, with a t-statistic of 91.43 and a p-value of 0.000.

- h. The correct statement is: “The univariate model underestimates the effect of being a top mover on the change in RH users. This is because top movers are smaller than the average company”.

6. The table below displays the estimated regression of the following model:

$$userchg = \beta_0 + \beta_1 lag_topmover + \beta_2 list + \beta_3 lag_topmover \cdot list + \beta_4 mktcapt + \varepsilon$$

. reg userchg lag_topmover list ltop_list mktcapt						
Source	SS	df	MS	Number of obs	= 1,742,651	
Model	5.8919e+09	4	1.4730e+09	F(4, 1742646)	= 14093.33	
Residual	1.8213e+11	1,742,646	104515.371	Prob > F	= 0.0000	
				R-squared	= 0.0313	
				Adj R-squared	= 0.0313	
Total	1.8803e+11	1,742,650	107896.12	Root MSE	= 323.29	
userchg	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
lag_topmover	138.6204	4.210243	32.92	0.000	130.3685	146.8723
list	19.03	.4915624	38.71	0.000	18.06655	19.99344
ltop_list	639.9109	6.257523	102.26	0.000	627.6463	652.1754
mktcapt	.0009139	5.72e-06	159.88	0.000	.0009027	.0009251
_cons	-4.834506	.3459101	-13.98	0.000	-5.512478	-4.156534

Table 7: regression lag_topmover on list, ltop_list, mktcapt

7. This model adds a dummy for the period after August 1, 2019, when Robinhood introduced the “Top Movers” feature, and an interaction with lagged top mover event. By also controlling for market capitalization, it isolates the effect of being a top mover from the extra impact of the new feature.

$$\beta_1 = 138.6204$$

The coefficient indicates that, holding other variables constant, a top mover event on the previous day is associated with an average increase of approximately 139 users before the first of August. The very small p-value (0.000) confirms that this effect is statistically significant at the 95% of confidence level.

$$\beta_3 = 639.9109$$

The coefficient measures how this effect (the effect previously described) changes after August 1st. It shows

that the impact of a top mover increases by an additional roughly 640 users after the first of August, when the list first appeared on the app. Thus, the total marginal effect of a top mover after August 1st is about 778 additional users. The extremely low p-value (0.000) here also confirms a highly statistically significance in the estimated coefficient.

Robinhood versus other investors

8.